A MORPHISM FROM AN AFFINE SCHEME IS QUASI-AFFINE EXACTLY WHEN IT IS QUASI-COMPACT

THE STACKS PROJECT AUTHORS

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Editorial note. Statement span 2358-2364 and proof span 2366-2373 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The statement and proof refer to the structure morphism of the base towards the spectrum of the integers in the auxiliary argument; the author's formulation is preserved.

Editorial note. 1 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

Allowed standard prerequisites (editorial).

- An open subscheme of an affine scheme is quasi-affine exactly when it is quasi-compact. Used at proof lines 2366-2369. Converts the condition on preimages of affine opens into quasi-compactness.
- Quasi-compactness of a morphism from an affine scheme to a quasi-separated scheme, deduced by applying the permanence statement to the composite with the structure morphism of the base, cited from the schemes chapter. Cited at proof lines 2370-2371. Gives the final assertion about quasi-separated targets.

Source reference. Original lemma 03GI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4529–4529. Same-chapter reference used in the retained author text at source line(s) 1003, 2371, 4298, 8937, 8952, 9829, 10100, 10769, 13984, 13991, 14143. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let S be a scheme. Let X be an affine scheme. A morphism $f : X \to S$ is quasi-affine if and only if it is quasi-compact. In particular any morphism from an affine scheme to a quasi-separated scheme is quasi-affine.*

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Proof. Let $V \subset S$ be an affine open. Then $f^{-1}(V)$ is an open subscheme of the affine scheme X, hence quasi-affine if and only if it is quasi-compact. This proves the first assertion. The quasi-compactness of any $f : X \to S$ where X is affine and S quasi-separated follows from Schemes, Lemma 03GI applied to $X \to S \to \mathrm{Spec}(\mathbf{Z})$. $\square$